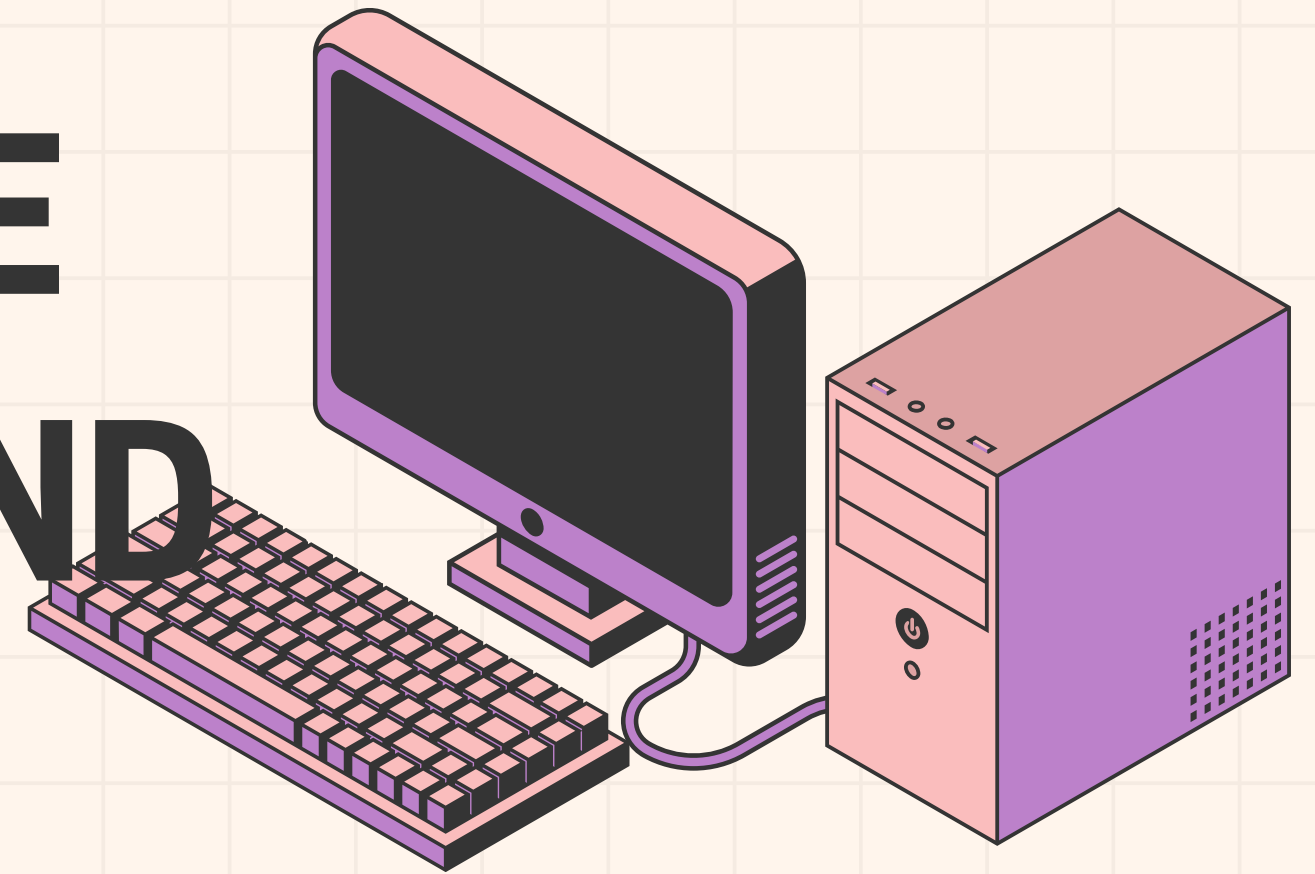


LESSON 4: ENTERPRISE RESOURCE PLANNING SYSTEMS AND BUSINESS PROCESS MODELS

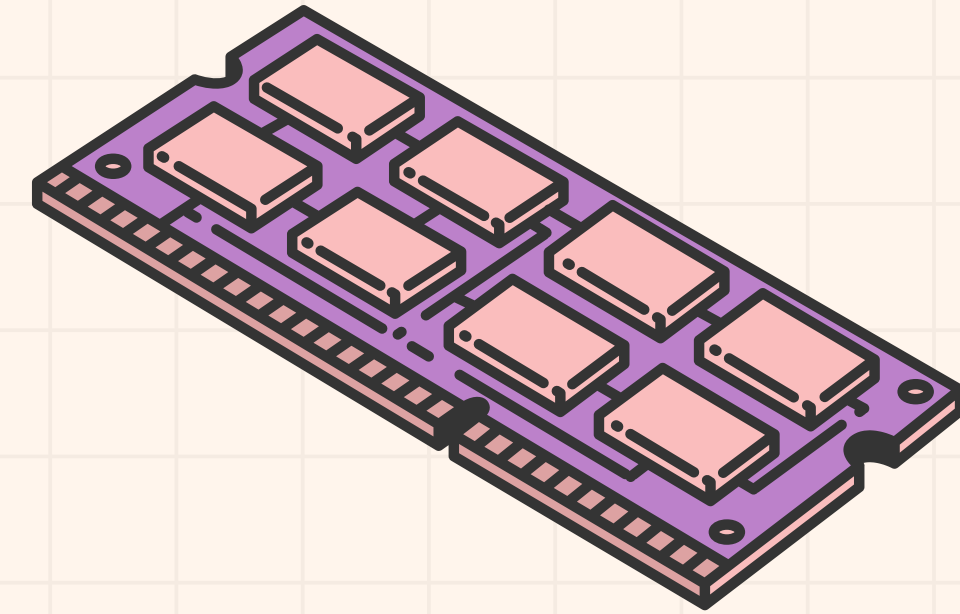


Enterprise Resource Planning (ERP) software is the backbone of most businesses, allowing them to manage data across all departments. It aids in the administration of company-wide business operations and the usage of a shared database and management reporting tools.

In this lesson, enterprise resource planning systems and business process models will be tackled to provide a broader perspective for students to deeply understand system integration.



LESSON OBJECTIVES



At the end of the lesson, the students must be able to

- 1. Identify the key role of ERP in business functions;**
- 2. Understand how ERP works in the functional areas of operation and business processes; and**
- 3. Define the functional areas of IS and Enterprise System Architecture**

Business processes are a set of actions that take an input and turn it into a useful output for the client.

An ERP system's primary job is to assist business functions.

To comprehend ERP, one must first comprehend how a firm operates.

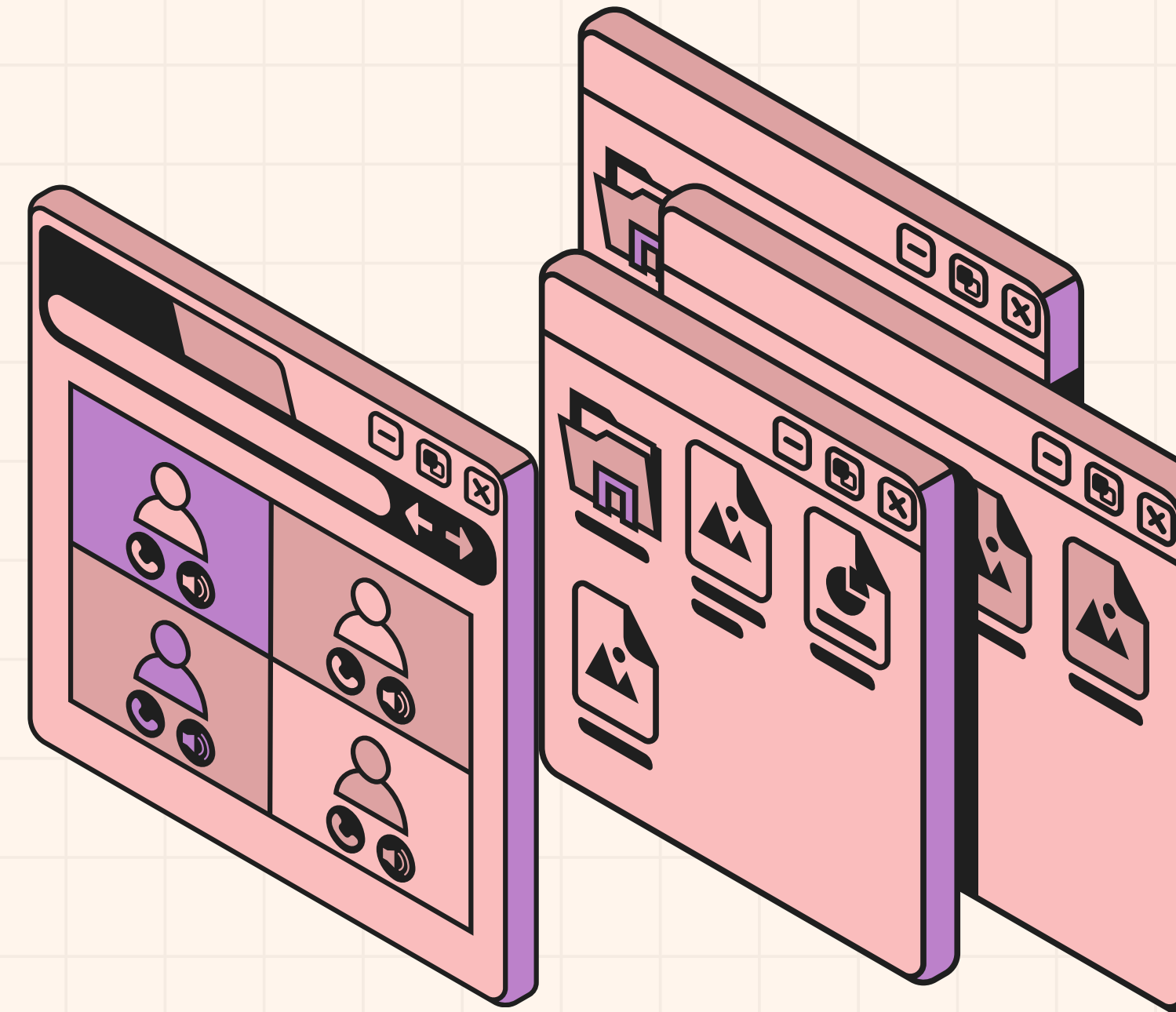
It's usually broken down into operational and business process functional categories.

Marketing and sales, supply chain management, accounting and finance, and human resources are all functional areas of operation.

Business functions, on the other hand, include activities specific to a functional area of operation.

ERP MODULES

ERP vendors such as SAP, Oracle, and Microsoft offer modules that support a company's major functional areas. The ERP software incorporates best business practices that use business rules to implement the organization's policies and procedures.



ERP MODULES

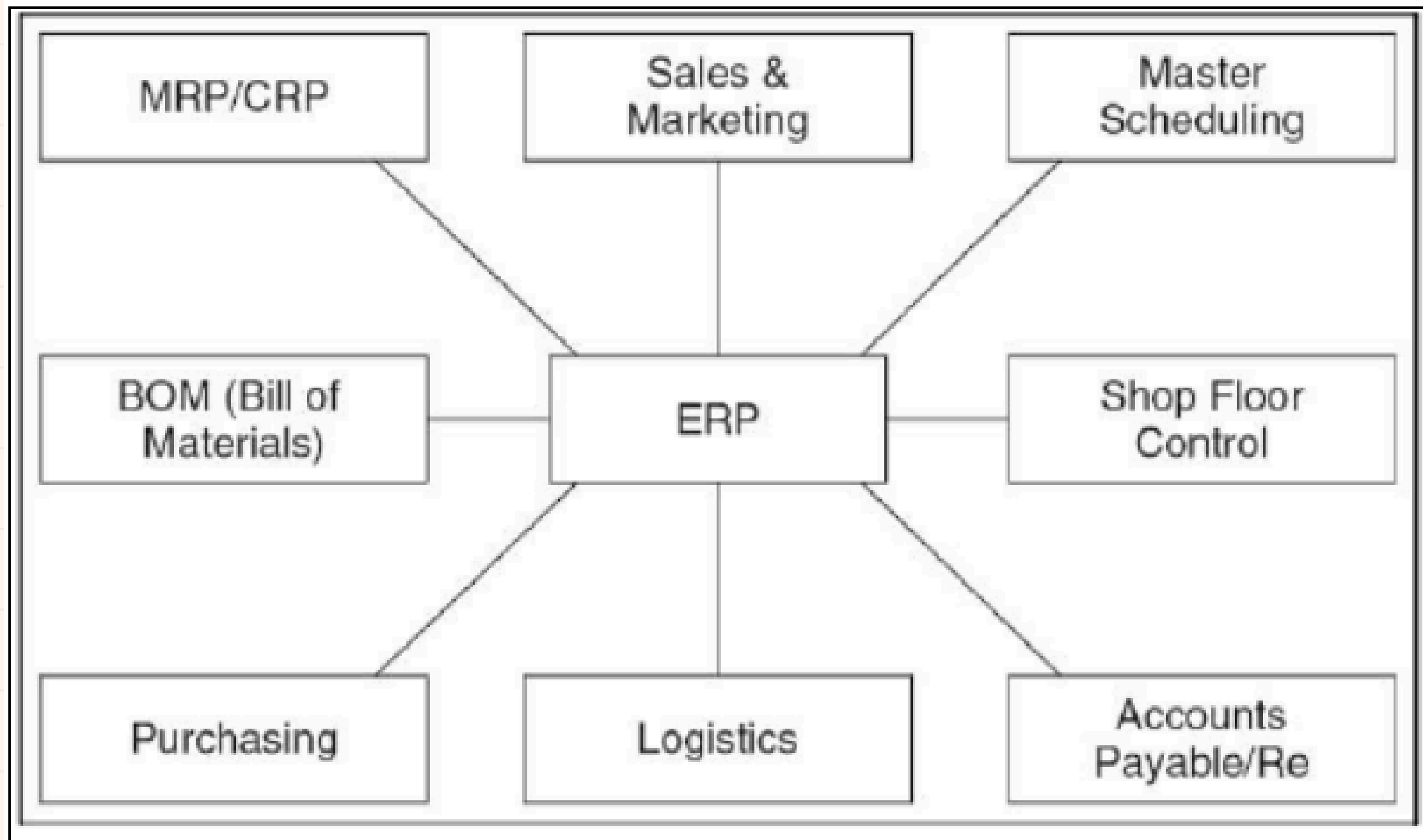
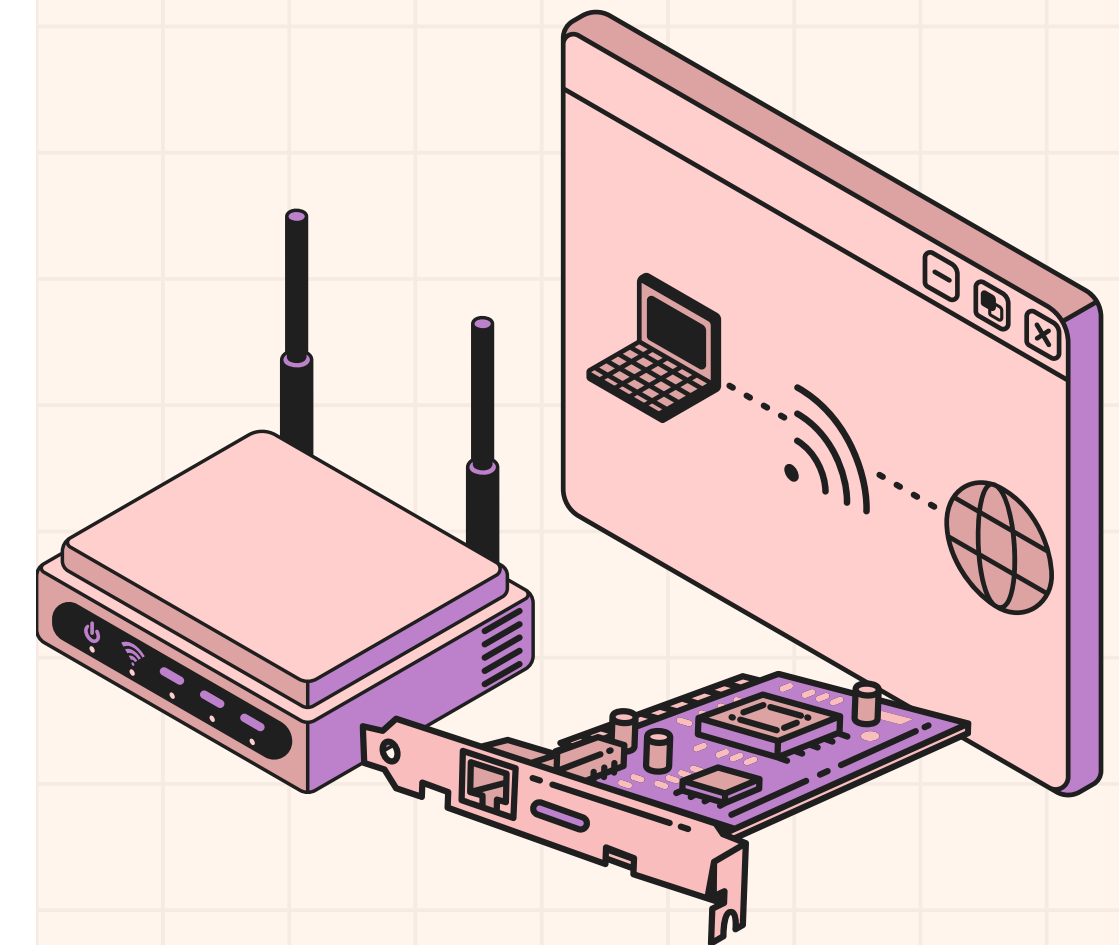


Figure 4-1: Example of ERP Modules

Image Source: Next Generation Application Integration Textbook

COMPARISON OF ERP MODULES FROM VENDORS

Function	SAP	Oracle/PeopleSoft	Microsoft Dynamics
Sales	Sales and Distribution, Sales Opportunity	Marketing and Sales, Supply Chain Management	Retail POS, Field Service Management
Procurement	Purchasing, Supplier Relationship Management	Procurement and Supplier Relationship Management	Supply Chain Management
Production	MRP, Product Life Cycle Management	Manufacturing	Manufacturing
Accounting	Financial Accounting	Financial Accounting	Financial Accounting
Distribution	Warehouse Management	Supply Chain Management	Distribution Management
Customer Service	CRM	CRM	CRM
Corporate Performance and Governance	Governance, Risk, and Compliance Management	Corporate Performance Management	Analytics
Human Resources	Human Capital Management	Human Capital Management	HR Management



HOW ERP MODULES WORK

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1. Marketing and Sales

Functions of marketing and sales module in an ERP includes advertising and marketing products, determining pricing, promoting products to 43 customer, taking customers' orders, helping create a sales forecast, customer support, and customer relationship management.

The Marketing and Sales module needs information from all other functional area. Customers communicate orders via Marketing and Sales in person or by telephone, e-mail, fax, the web, etc.

2. Supply Chain Management

The functions within supply chain management includes making the product (manufacturing/production), buying raw materials(purchasing), receiving goods and raw materials, transportation and logistics, scheduling production of products, and plant maintenance.

HOW ERP MODULES WORK

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3. Accounting and Finance

Accounting and finance module includes recording raw data about transactions (including sales), raw materials purchases, payroll, and receipt of cash from customers, planning & budgeting, cost allocation & control.

4. Human Resources

Human resources functions include recruit, train, evaluate, and compensate employees, benefits, and government compliance. HR uses data from other modules to plan personnel development path. Systems integrated using ERP software provide the data sharing necessary between functional areas.

5. Production

Helps in the planning and optimizing of the manufacturing capacity, parts, components, and material resources using historical production data and sales forecasting.

HOW ERP MODULES WORK

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6. Purchasing

Streamlines the procurement process of required raw materials and other supplies

7. Inventory Management

Facilitates the processes of maintaining the appropriate level of stock in a warehouse

8. Miscellaneous Modules

Nontraditional modules such as business intelligence, self-service, project management, and e-commerce.

COMPONENTS OF THE ENTERPRISE SYSTEMS ARCHITECTURE

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a. Functional – Defines the ERP modules that support the various business functions of the organization. Examples include Accounting, Human Resources, Procurement, etc.

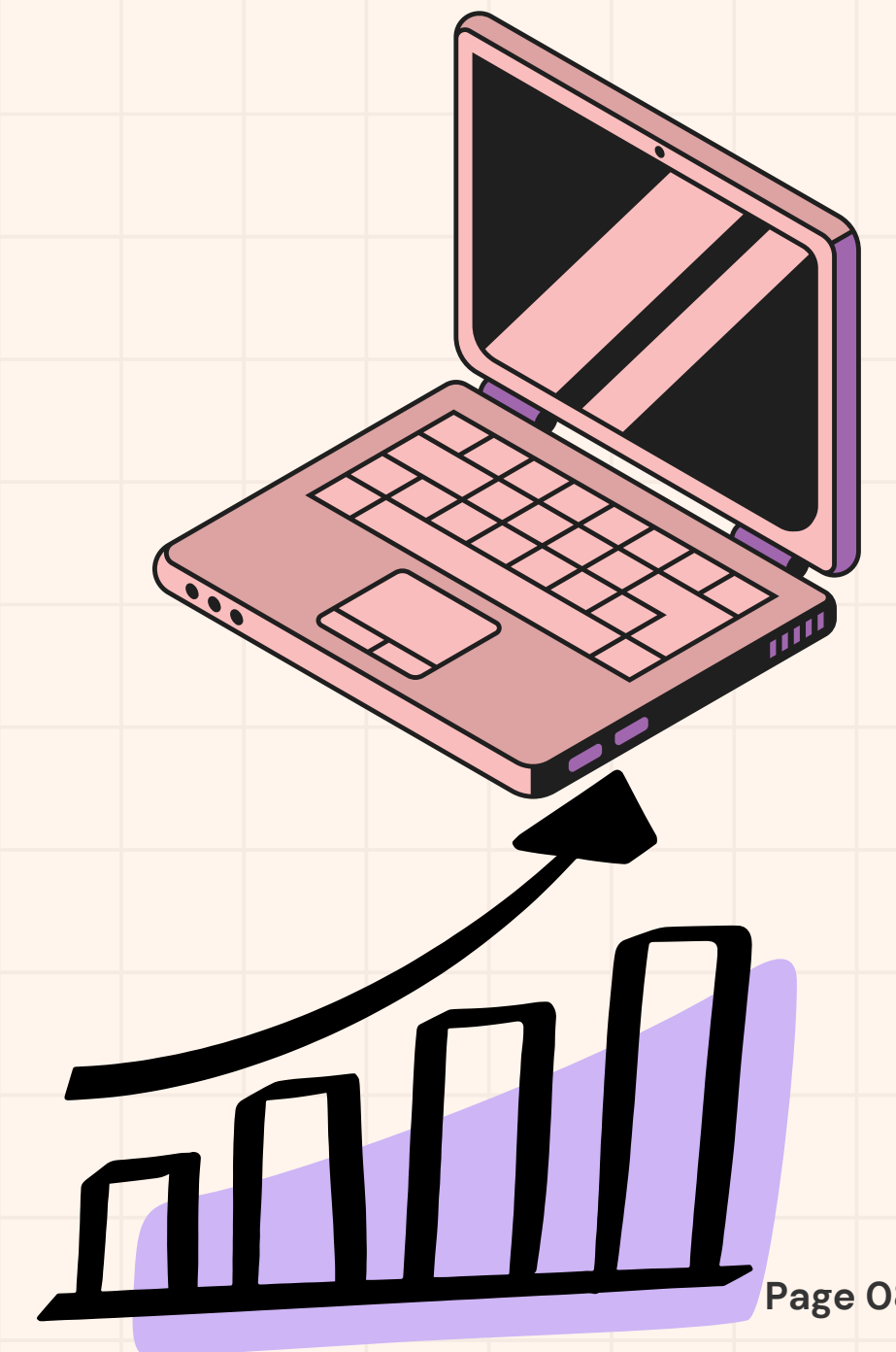
b. System – Defines the ERP Architecture through the physical components of hardware, software, and networking angle.

ENTERPRISE SYSTEMS ARCHITECTURE

The architecture aids implementation teams in fully comprehending the enterprise system's features and components.

It depicts the intricate system interfaces that exist between the ERP program and databases, operating systems, legacy applications, and networking.

If the needs for system infrastructure, training, change management, and business process reengineering are clear, management may design a stronger IT plan.



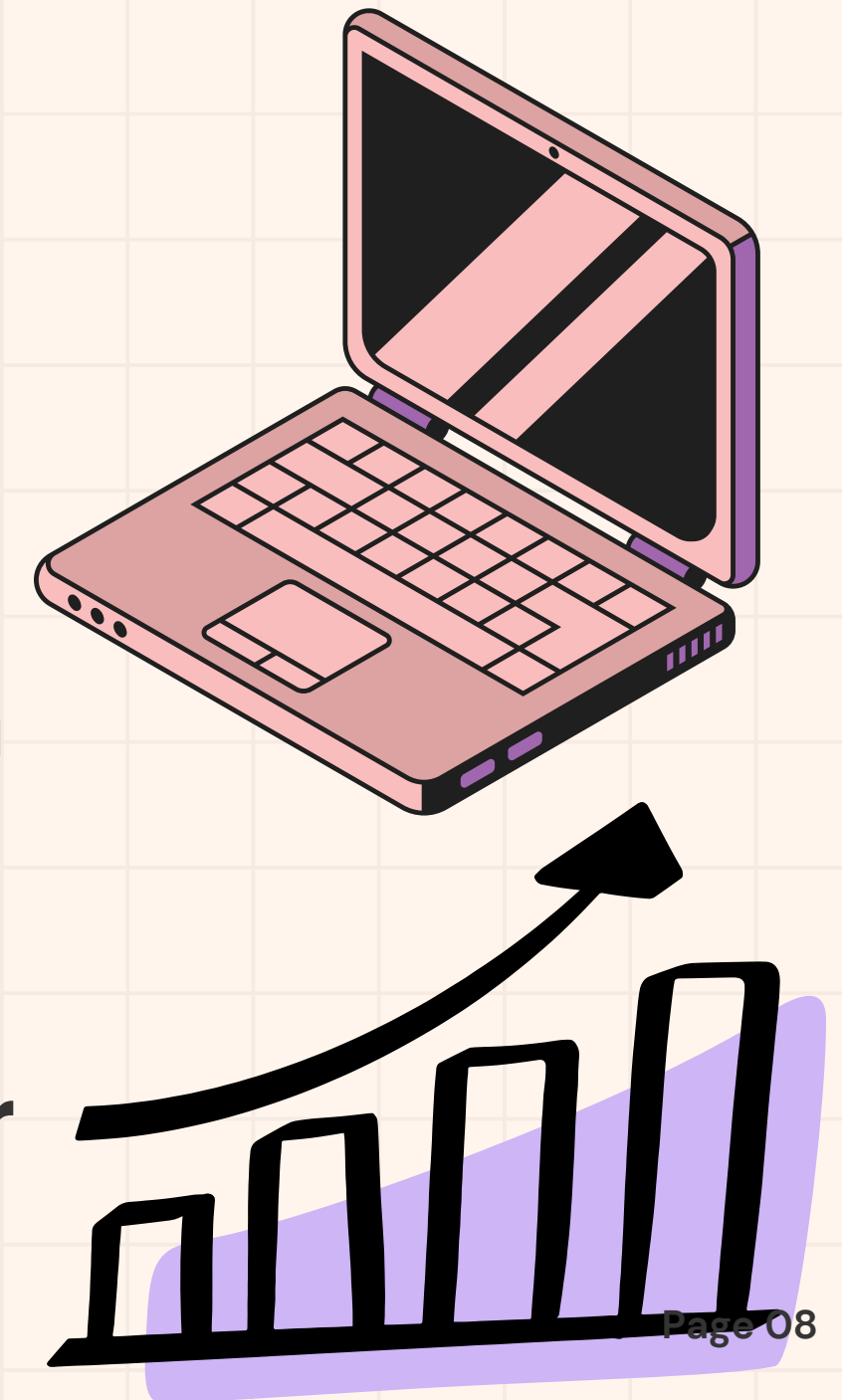
THREE-TIER ARCHITECTURE

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A three-tiered design, consisting of a Web Tier, an Application Tier, and a Data Tier, is used in the majority of modern ERP installations.

Benefits include:

- 1. Scalability – Easier to add, change, and remove applications.**
- 2. Reliability – Implementing multiple levels of redundancy**
- 3. Flexibility – Flexibility in partitioning is very simple**
- 4. Maintainability – Support and maintenance costs are less on one server**
- 5. Reusability – Easier to implement reusable components**
- 6. Security – IT staff has more control system to provide higher security**



TIERS

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1. The Web Tier

Web-based portal allows users the ability to access and analyze information through their web browser.

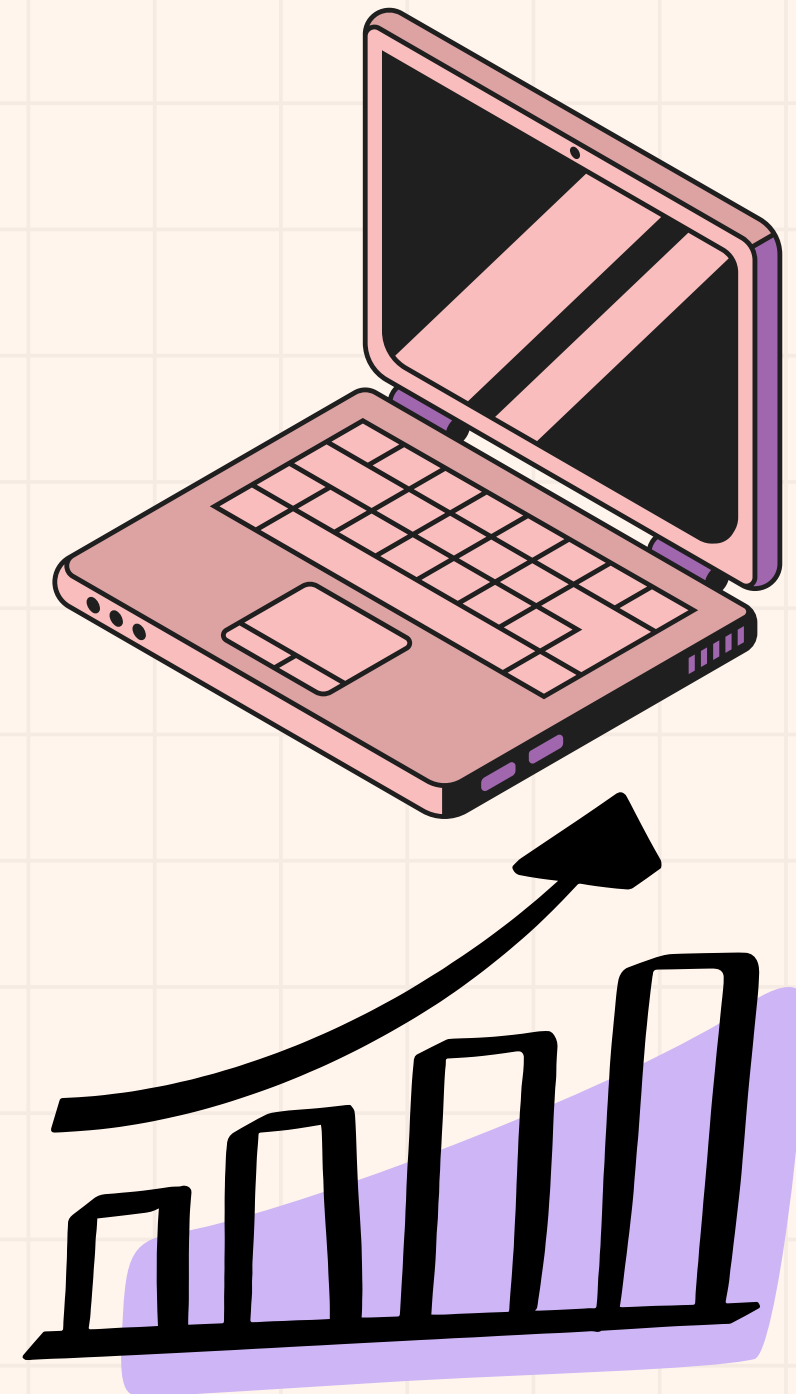
2. The Application Tier

The application tier consists of a web browser and reporting tool where business processes and end-users interact with the system.

It shields the business users from the inner workings of an ERP system, but still provides the information relevant to their job and business process.

3. The Data Tier

Data tier focuses on structure of all organizational data and its relationship with both internal and external systems.



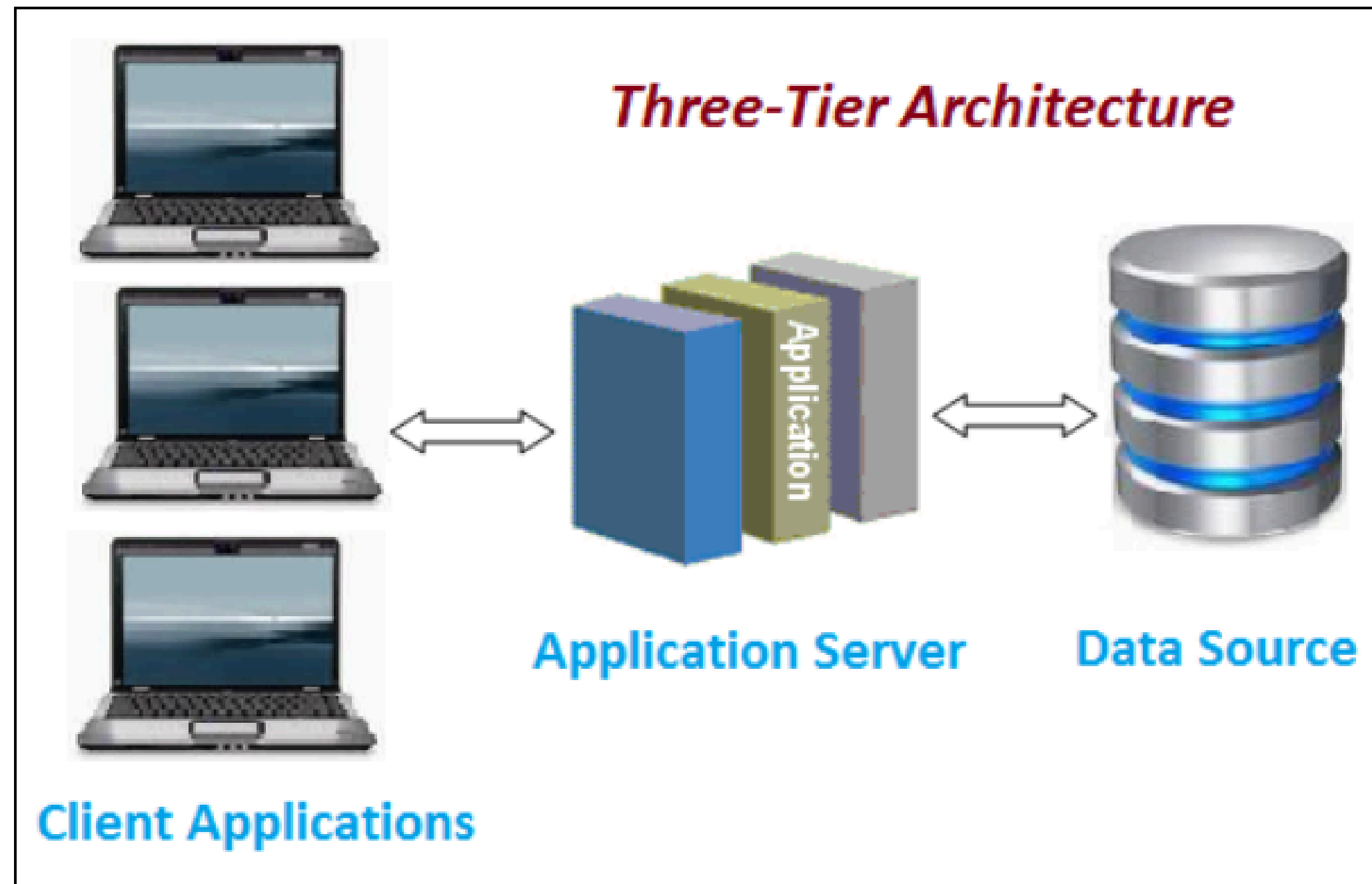
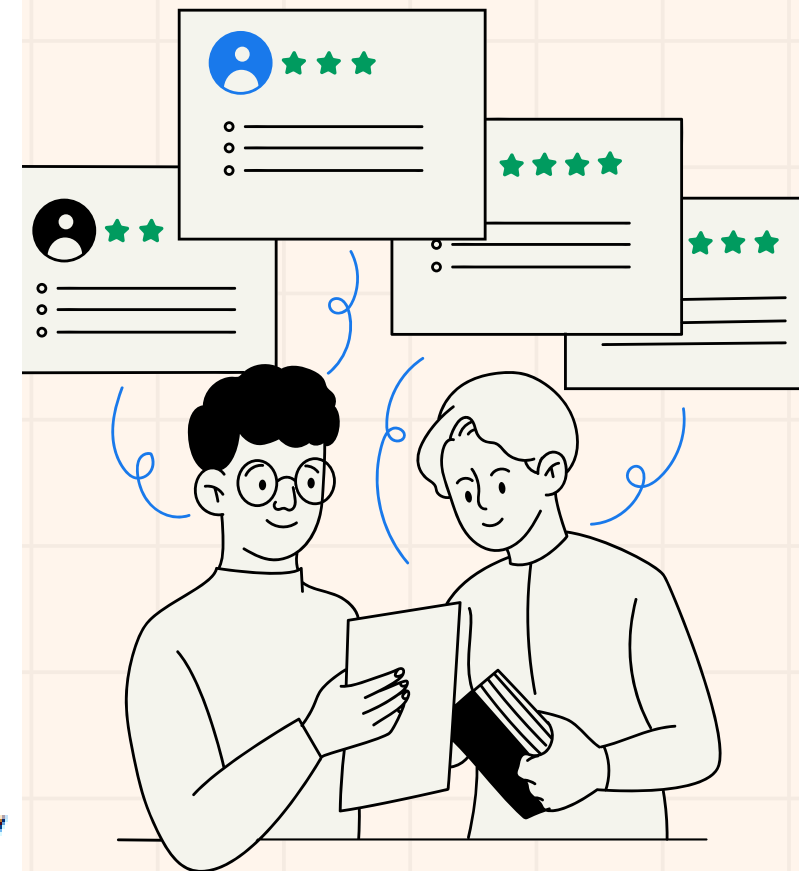


Figure 4-2: Three Tier Architecture

Image Source: <https://www.softwaretestingclass.com/what-is-difference-between-two-tier-and-three-tier-architecture/>

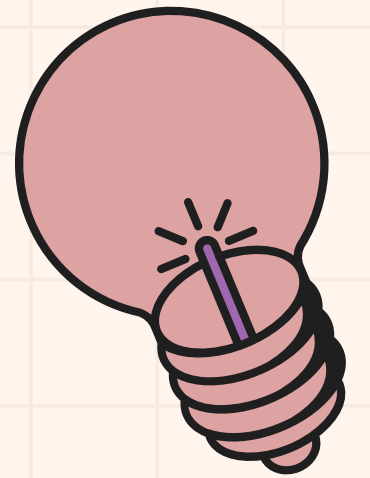
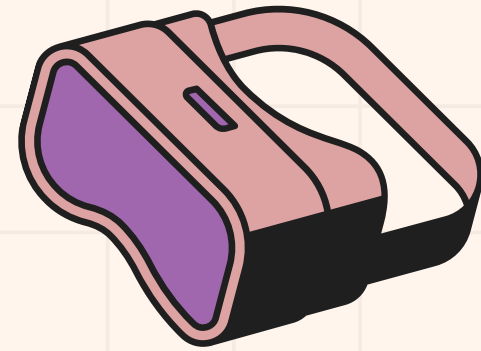


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THANK YOU

